Predicting Order-Disorder Phase Transitions in the Vicsek Flocking Model

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Introduction

The Vicsek model, proposed in 1995 by Tamás Vicsek, is a foundational agent-based model created to try to capture the unique behavior of collective motion, the spontaneous movement of many self-propelled particles. Proposed as a bird flocking model, the Vicsek model depicts the onset of collective motion from complete initial disorder.

In this work, we use machine learning as a tool through to investigate the phase transitions in the agent- based Vicsek model to both elucidate the controversial phase boundaries in the Vicsek model and demonstrate the utility of machine learning in complex agent-based collective motion models.

Model Overview

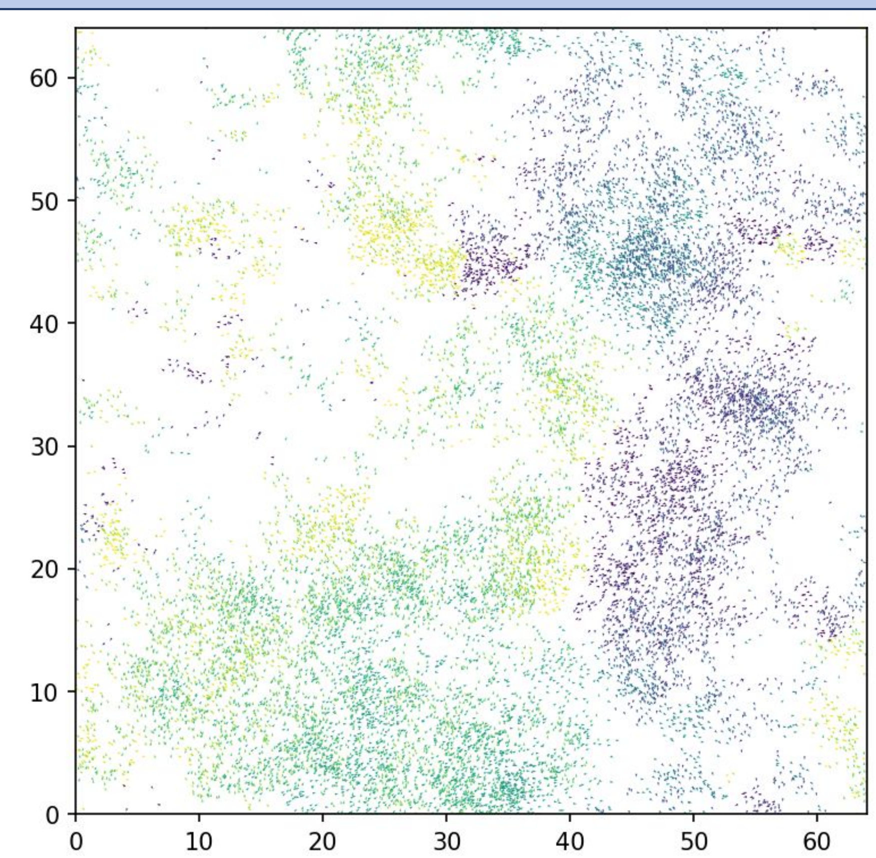
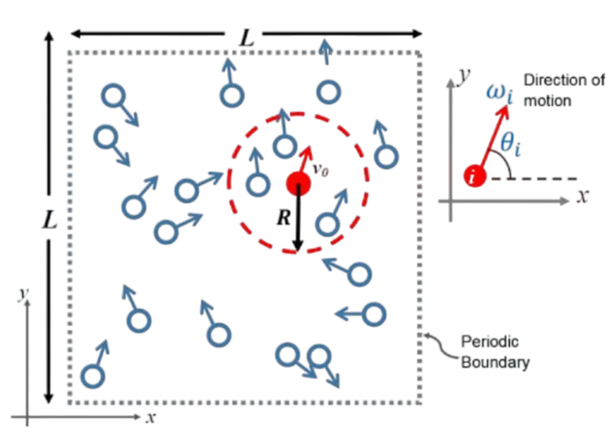


FIG. 1. Snapshot of the animated Vicsek model at 1,000 time steps with L = 64, $\eta = 0.3$, $\rho = 4$, and $v_0 = 0.1$. The particle color spectrum represents the direction of motion, where particles of the same color are moving in the same direction and particles of different colors are moving in different directions.

The Vicsek model defines N active self-propelled particles moving at a constant speed v_0 in 2D space with linear size L.

$$\theta(t + \Delta t) = \langle \theta(t) \rangle_r + \Delta\theta$$

particle direction = avg. direction of neighbors + noise



The only rule: At each time step, each particle moves in the average direction of motion of its neighbors + a small amount of noise as seen in the formula and picture above.

Each simulation (Fig. 1) reaches a final state, or phase, of order, coexistence, or disorder from initial disorder depending on the free parameters **noise η**, **density ρ**, and **speed v_0**. We can thus analyze order-disorder phase transitions as a function of these free parameters.

Methodology

To effectively detect phase transitions in the Vicsek model, we want to establish a clean global picture of where the different collective states exist by utilizing the methodology below:

Selection of parameter points ⇨ Construction of labeled dataset using chosen parameter points ⇨ Training of a neural network on dataset to determine phase boundaries

Simulation: In this study, a numbers-only C++ implementation of the Vicsek model is used with TJ's Supercomputer Cluster to optimize computational efficiency. All parameters are kept constant across runs except the free variables η, ρ, and v_0 with a consistent system size of L = 128. Each simulation is run to 1,000,000 time steps to ensure final-state convergence. The simulation metrics of order parameter ϕ, stationary order parameter $\langle\phi\rangle$, and Binder Cumulant G are calculated as:

$$\phi(t) = \frac{1}{N v_0} \left| \sum_{i=0}^{N} v_i(t) \right|, \quad \langle \phi \rangle = \lim_{T \to \infty} \frac{1}{T} \int_0^T \phi(t) dt, \quad G = 1 - \frac{\langle \phi^4 \rangle}{3 \langle \phi^2 \rangle^2}$$

where ϕ measures the order of a simulation at a specific point in time, $\langle\phi\rangle$ measures the steady-state order of a simulation, and G measures the fluctuation of order in the steady state.

Dataset Construction: We use Latin Hypercube Sampling (LHS), a form of stratified random sampling, to select an even distribution of parameter points for dataset construction. For each chosen parameter point, $\lambda = (\eta, \rho, v_0)$, we run 5 simulation trials with those parameters and random initial conditions.

Each parameter point, or dataset instance, is labeled with the K-Means clustering algorithm in $(\langle\phi\rangle, G, \chi)$ space with 3 centroids, where the formulas for $\langle\phi\rangle$ and G are seen above, and susceptibility χ is calculated with

$$\chi = \langle (\phi^2 - \langle \phi \rangle^2) \rangle$$

where χ similarly measures the fluctuation of order in the steady state. Using these methods, we thus build a dataset with the 3 independent features in (η, ρ, v_0) space, 193 instances, and 3 nominal labels representing the order, banding and disorder phases.

Classification: After assigning labels to the dataset, we use a fully connected feed-forward neural network trained with a stratified train-test split and cross-entropy loss, where each input is represented by the 3D feature vector $\mathbf{x} = (\eta, \rho, v0)$, to classify points on a grid by interpolating between the simulated data points to produce a phase diagram.

Results

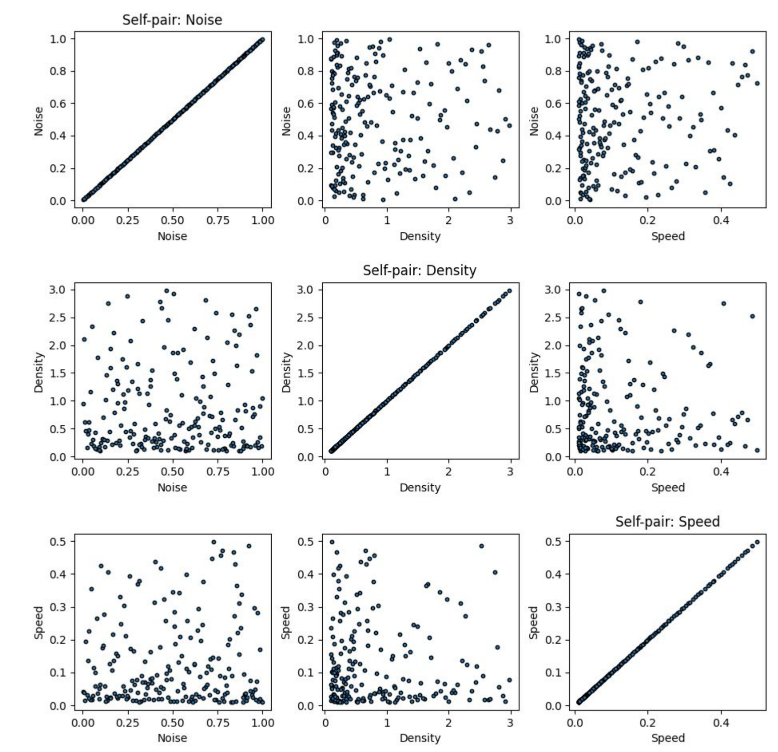


FIG. 2. Stratified Latin Hypercube Sampling results in two-dimensional space with N = 200, noise η on a linear scale, and density ρ and speed v_0 on a $\log_{10}$ scale.

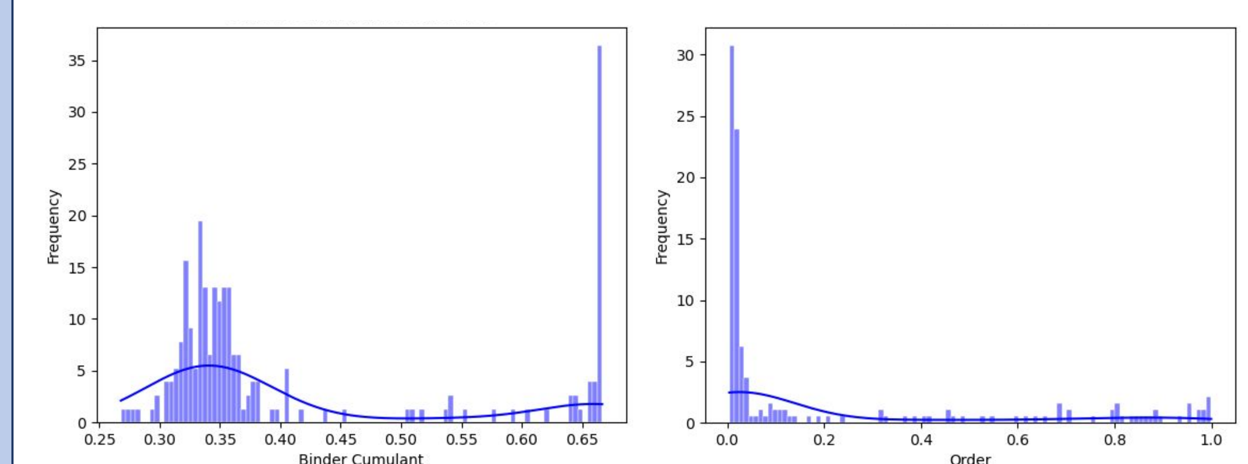


FIG. 3. Simulation metric frequency distributions over 193 simulation trials. (a) Binder Cumulant G frequency distribution with a peak at G = ⅓ and G = ⅔. (b) Order parameter ϕ frequency distribution over the same simulations.

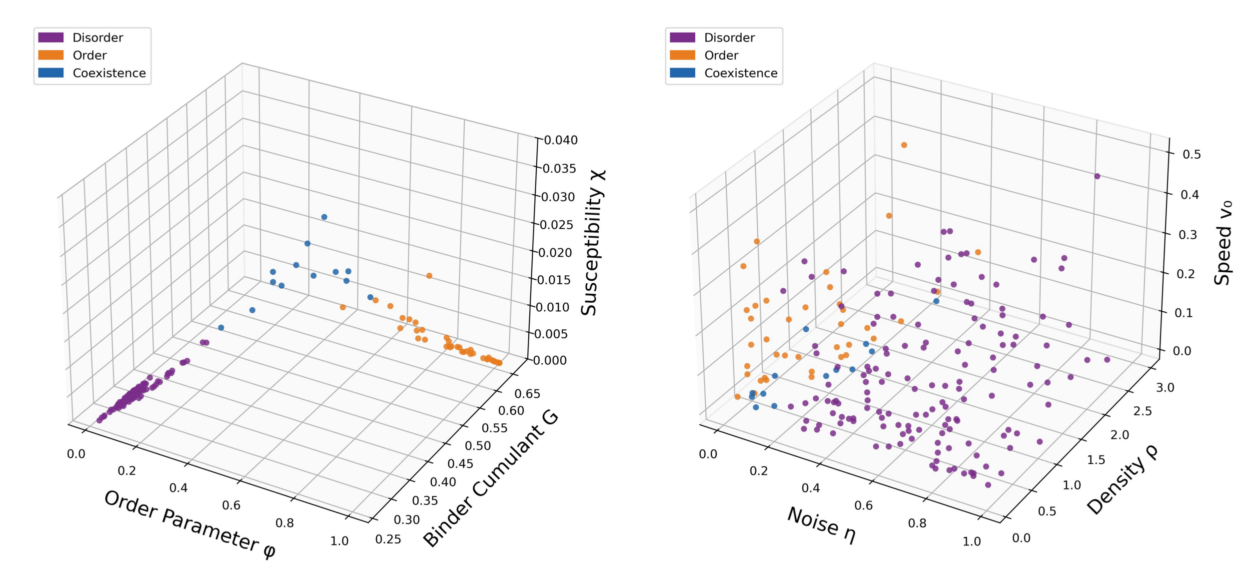


FIG. 4. Three-dimensional view of K-Means classification of Vicsek-model simulation data. (a) Clustering in the observable space spanned by the order parameter $\langle\phi\rangle$, Binder cumulant G, and susceptibility χ. (b) The same cluster labels projected onto the parameter space (η, ρ, v_0), showing the corresponding disorder, order, and coexistence regions.

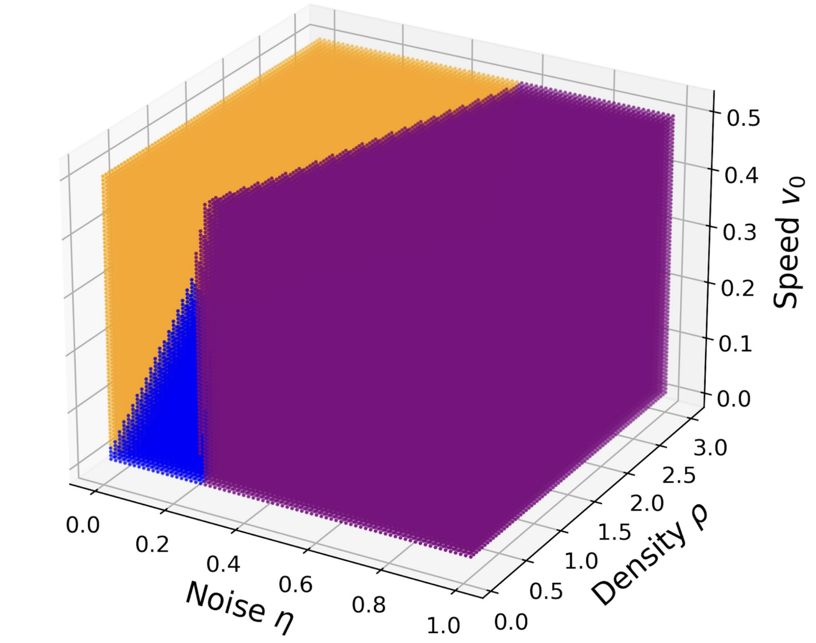


FIG. 5. Three-dimensional learned phase diagram of the Vicsek model in the parameter space (η, ρ, v0). Each point represents a classified parameter-grid point, with purple, orange, and blue denoting disorder, order, and coexistence, respectively. The classifier predicts a broad ordered region at low noise, a disordered region at higher noise, and a coexistence region separating the two phases.

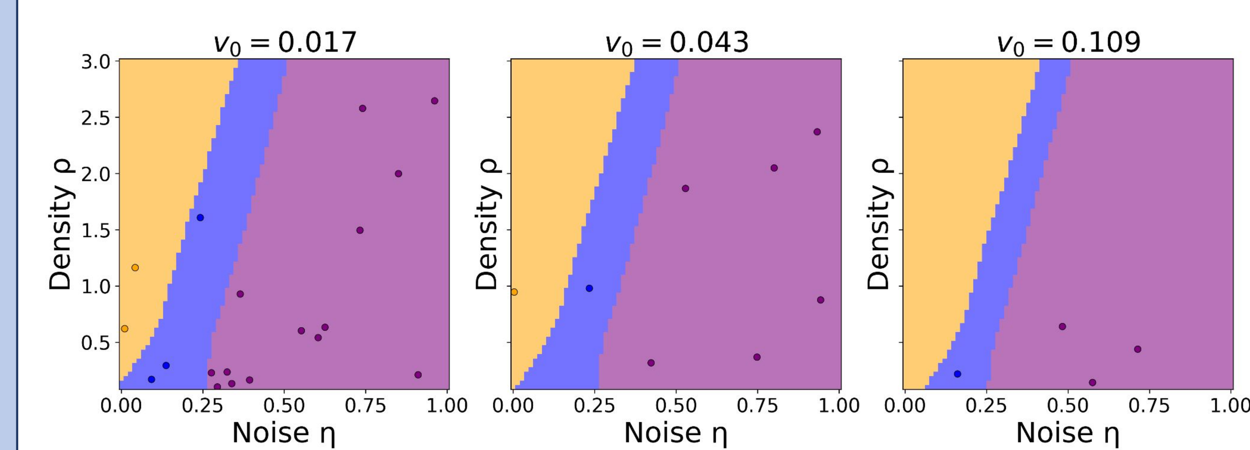


FIG. 6. Two-dimensional slices of the learned Vicsek-model phase diagram in the (η, ρ) plane at fixed particle speeds v0 = 0.017, 0.043, and 0.109. Shaded regions show neural-network classification on an evenly spaced parameter grid, while overlaid points show the original labeled simulation data at each slice. Purple, orange, and blue denote disorder, order, and coexistence, respectively.

We produce a global coarse phase map in (η, ρ, v_0) space with the help of the neural network and labeled dataset (Figs. 2-4), where the neural network assigns a nominal label to each instance with an accuracy of 0.92.

In these results, we can observe the effect of the free variables η, ρ, and v_0 on the final dominant phase of each simulation, as seen above from the phase boundary lines in Figs. 5 and 6. While η appears to be the greatest contributing parameter to phase transitions, we can note that varying v_0 separates the ordered and coexistence phases, with ρ affecting the boundary between the ordered and disordered phase at high values of v_0.

Conclusion and Future Directions

In this work, we used machine learning as a tool to construct a phase diagram of the Vicsek model in (η, ρ, v_0) space with the K-Means clustering algorithm and an artificial neural network. This research produces results that provide more insight into phase transitions in the Vicsek model and validate the efficacy of applications of machine learning to collective motion models. These results also introduce potential of machine learning in phase transition areas in the Vicsek model provided by the phase map produced in this work.

References

Demo Video

Full Paper